LAB-ASSIGNMENT-2

NAME:- Ankit kumar

ENROLLMENT N0:- S25MCAG0010

1). **Display the number of buses from each source**

```
mysql> SELECT Source, COUNT(*) AS Total_Buses
    -> FROM List_Buses
    -> GROUP BY Source;
+------------+-------------+
| Source     | Total_Buses |
+------------+-------------+
| Pari Chauk |           3 |
| Sector 62  |           1 |
| ISBT       |           4 |
+------------+-------------+
3 rows in set (0.00 sec)
```

2).Display the number of buses from each source, sorted low to high.

```
mysql> SELECT Source, COUNT(*) AS Total_Buses
    -> FROM List_Buses
    -> GROUP BY Source
    -> ORDER BY Total_Buses ASC;
+------------+-------------+
| Source     | Total_Buses |
+------------+-------------+
| Sector 62  |           1 |
| Pari Chauk |           3 |
| ISBT       |           4 |
+------------+-------------+
3 rows in set (0.01 sec)
```

3). Display all the details of first 4 cheapest buses from list_buses table.

```
mysql> SELECT *
    -> FROM List_Buses
    -> ORDER BY Fare ASC
    -> LIMIT 4;
+--------+-------------+------------+-------------+------+------------+----------+
| bus_no | bus_name    | source     | destination | fare | date       | time     |
+--------+-------------+------------+-------------+------+------------+----------+
| DL306  | Shatabdi    | ISBT       | Dehradun    | 1007 | 2022-09-29 | 20:45:00 |
| UP301  | Zing bus    | Pari Chauk | Prayagraj   | 1010 | 2022-08-21 | 21:30:00 |
| HR303  | Maheshwaram | ISBT       | Ambala      | 1010 | 2022-09-25 | 07:22:00 |
| HR302  | Cargo       | Sector 62  | Chandigarh  | 1020 | 2022-08-23 | 22:44:00 |
+--------+-------------+------------+-------------+------+------------+----------+
4 rows in set (0.00 sec)
```

4). Display all the passengers whose name begins with D.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE P_Name LIKE 'D%';
+--------------+--------+--------+--------+------+---------+
| passenger_id | p_name | bus_no | gender | age  | status  |
+--------------+--------+--------+--------+------+---------+
| SCS018       | Divya  | UP101  | F      |   65 | Waiting |
| SCS014       | Diya   | HR303  | F      |   19 | Confirm |
+--------------+--------+--------+--------+------+---------+
2 rows in set (0.00 sec)
```

5). Write a query that selects the first passenger in alphabetical order whose name ends with m.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE P_Name LIKE '%m'
    -> ORDER BY P_Name ASC
    -> LIMIT 1;
+--------------+--------+--------+--------+------+---------+
| passenger_id | p_name | bus_no | gender | age  | status  |
+--------------+--------+--------+--------+------+---------+
| SCS022       | Alam   | HR302  | M      |   21 | Waiting |
+--------------+--------+--------+--------+------+---------+
1 row in set (0.00 sec)
```

6). Select the average age of male passengers.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE P_Name LIKE '%m'
    -> ORDER BY P_Name ASC
    -> LIMIT 1;
+--------------+--------+--------+--------+------+---------+
| passenger_id | p_name | bus_no | gender | age  | status  |
+--------------+--------+--------+--------+------+---------+
| SCS022       | Alam   | HR302  | M      |   21 | Waiting |
+--------------+--------+--------+--------+------+---------+
1 row in set (0.00 sec)
```

7). Display the name of the Passenger in Confirm status, with maximum age.

```
mysql> SELECT P_Name
    -> FROM Passengers
    -> WHERE Status="Confirm"
    -> ORDER BY Age DESC
    -> LIMIT 1;
+----------+
| P_Name   |
+----------+
| Abhishek |
+----------+
1 row in set (0.00 sec)
```

8). Display all the students who are female and have confirm tickets and sourceid from pari chauk.

```
mysql> SELECT *
    -> FROM passengers
    -> WHERE Gender = 'F'
    -> AND Status = 'Confirm'
    -> AND Bus_no IN (
    ->     SELECT Bus_no
    ->     FROM List_buses
    ->     WHERE Source = 'Pari Chauk'
    -> );
+--------------+--------+--------+--------+------+---------+
| passenger_id | p_name | bus_no | gender | age  | status  |
+--------------+--------+--------+--------+------+---------+
| SCS043       | Rupam  | UP501  | F      |   22 | Confirm |
+--------------+--------+--------+--------+------+---------+
1 row in set (0.04 sec)
```

9). Combine Age and Gender column as of passenger_detail column.

```
mysql> SELECT CONCAT(Age, '-', Gender) AS Age_Gender
    -> FROM Passengers;
+------------+
| Age_Gender |
+------------+
| 17-M       |
| 15-F       |
| 65-F       |
| 19-F       |
| 57-M       |
| 17-M       |
| 21-M       |
| 22-F       |
| 23-F       |
| 21-M       |
| 23-M       |
+------------+
11 rows in set (0.02 sec)
```

10). Write a SQL statement to increase the fare of buses from Pari Chauk and ISBT. The fare will be increased by 50 and 75 for Pari Chauk and ISBT respectively and the fare from for rest of the sources will remain same.

```
mysql> UPDATE List_Buses
    -> SET Fare =
    -> CASE
    ->     WHEN Source = 'Pari Chauk' THEN Fare + 50
    ->     WHEN Source = 'ISBT' THEN Fare + 75
    ->     ELSE Fare
    -> END;
Query OK, 7 rows affected (0.07 sec)
Rows matched: 8  Changed: 7  Warnings: 0
```